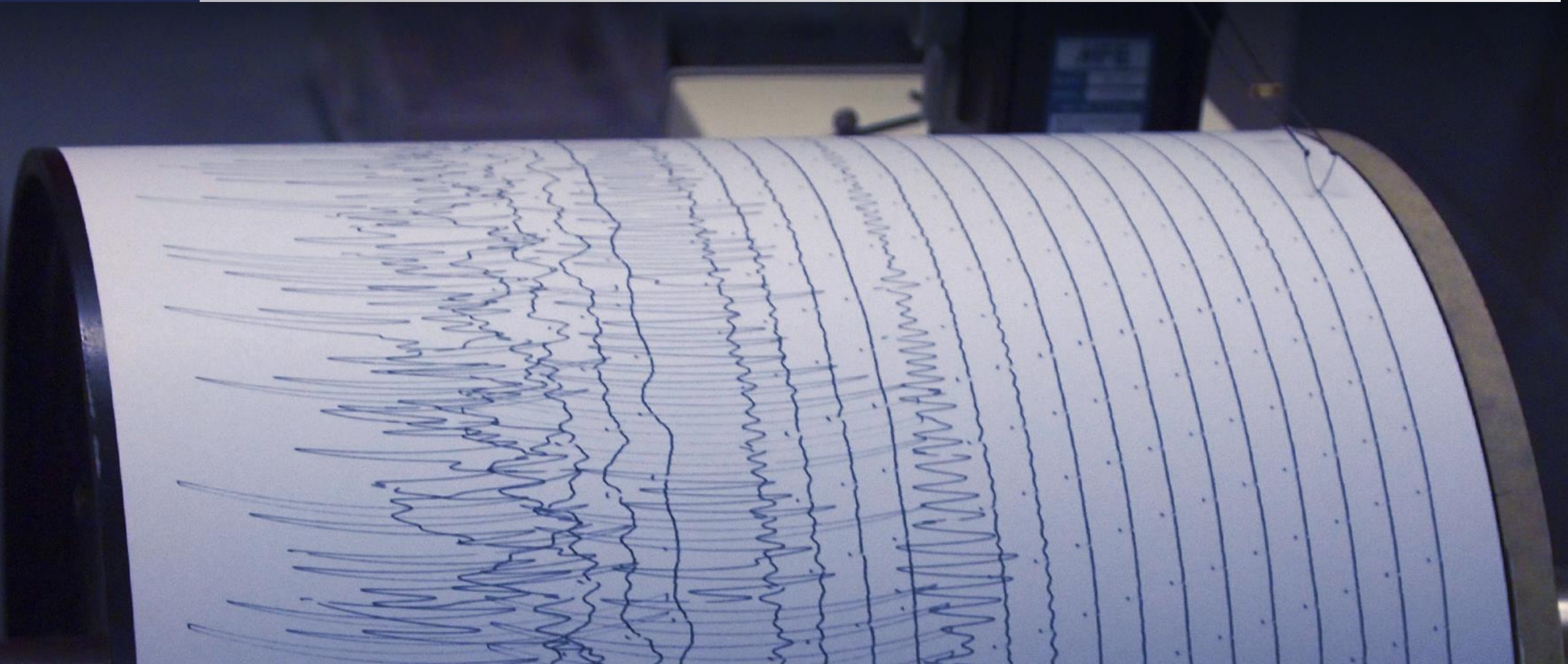


$f(x)$



Calculus lecture 1

Functions and continuity



Assessment

- Week 2.5: short answer test 1
- Week 2.10: open question test 2
- Passed if

$$\frac{\text{test 1} + 2 \cdot \text{test 2}}{3} \geq 5.75 \quad \text{and} \quad \text{test 2} \geq 5$$

Single resit (100%), Highest grade counts.

For more information:

Brightspace (enroll in the course **CSE1200 Calculus**) and see the course organization document under course documents.

Questions?

course organization document

Organization of the course is laid out in detail in the "course organization document" found in Brightspace under **content>course documents**.

answers

Questions can be asked through:

<https://answers.ewi.tudelft.nl>

Make sure to include the tag **cse1200** and to search for duplicate questions before you post.

E-mail

For questions that are not appropriate for the forum use the dedicated course e-mail: **CSE1200@tudelft.nl**

Programme

- Functions
- Inverse functions
- Inverse trigonometric functions
- The limit of a function
- Continuity



Functions

Definition (Function)

A *function* from A to B , denoted as $f : A \rightarrow B$, is a *relation* from A , the *domain*, to B , the *co-domain*, that satisfies two conditions:

- 1 *every* element of A is related to some element of B ;
- 2 no element of A is related to more than one element of B .

That is to say, a function f relates *every element* x from the *domain* to a *unique element* $f(x)$ from the *co-domain*.

A function $f : A \rightarrow B$ that violates at least one of these conditions is **not** *well-defined*.

Single Variable Calculus

For us a function

$$f: A \longrightarrow B$$

has as domain A and co-domain B a subset of the real numbers.

The **range** of f is the set of real numbers that are reached by f .



Polling question using Vevox



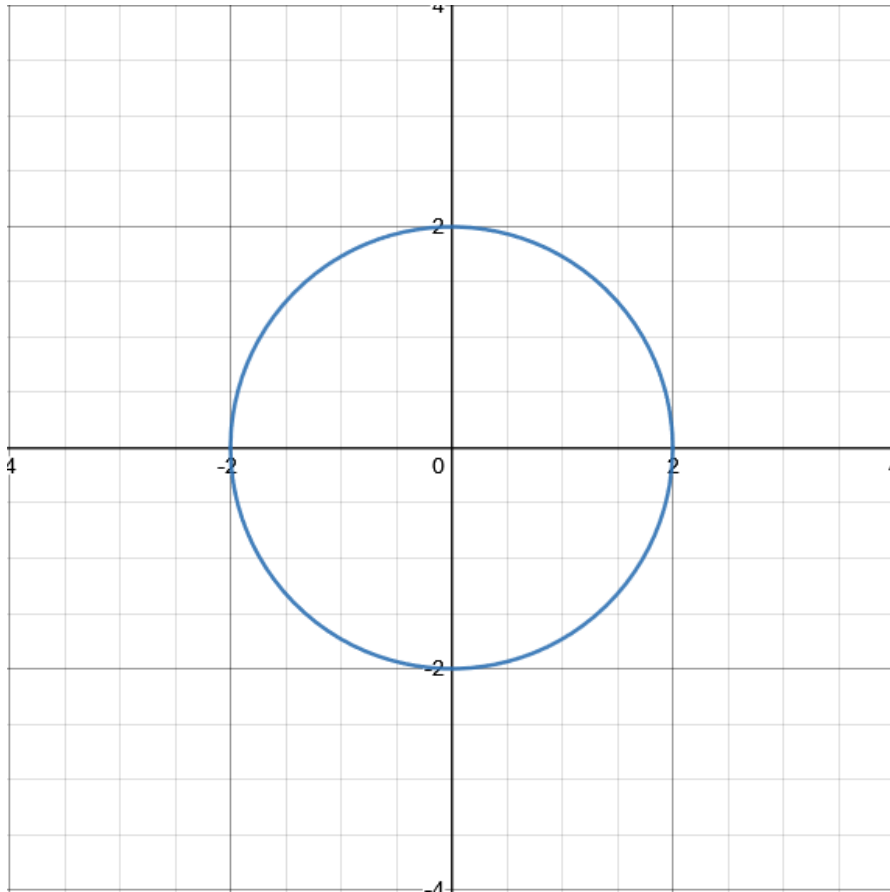
1. Go to Vevox.app and enter the session ID or scan the QR code.
2. Wait for a question to appear



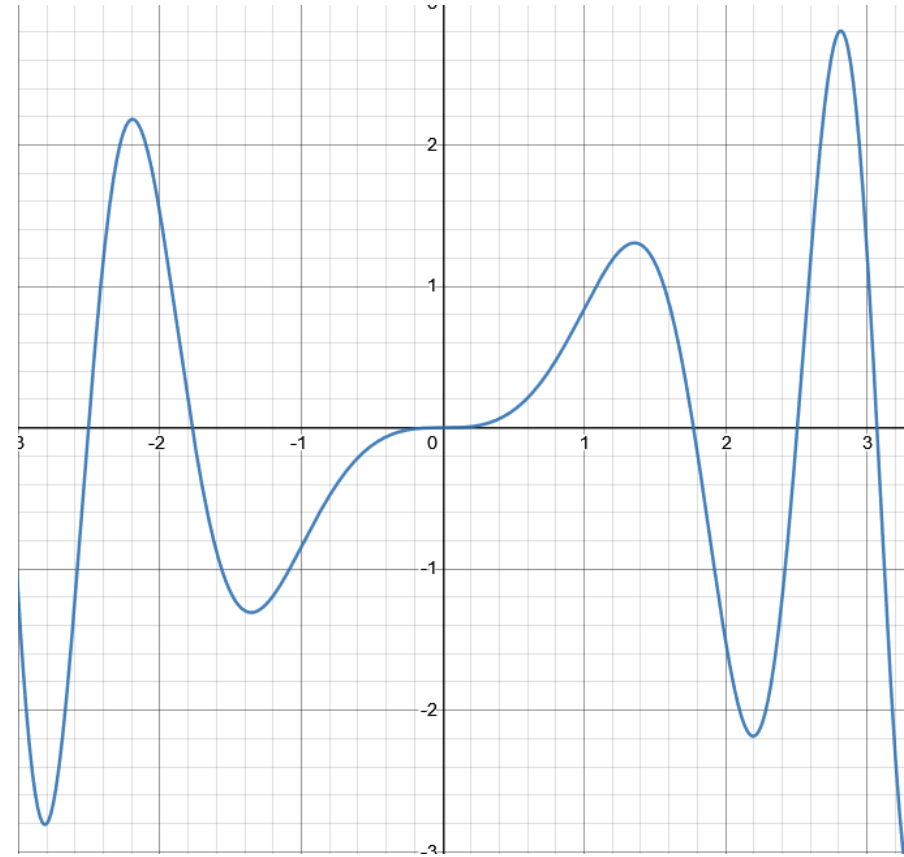
Graph of a function?

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No



Yes



Product and Composition

The product of two functions $f(x)$ and $g(x)$ is again a function.

$$h(x) = f(x)g(x)$$

The composition of two functions $f(x)$ and $g(x)$ is again a function

$$h(x) = (f \circ g)(x) = f(g(x))$$



Inverse functions

Inverse functions

Definition:

Suppose f is a function with domain D and range R .

A function g is the inverse of f (on R)

if for all x in D and y in R the following holds:

$$f(x) = y \iff x = g(y)$$

Notation inverse function: f^{-1} .

Note that $f(f^{-1}(y)) = y$ and $f^{-1}(f(x)) = x$.



Find the inverse

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Let $f(x) = 3x + 5$ with domain $\mathbb{R}$ and with codomain $\mathbb{R}$.

- A. The inverse of f is $g(x) = \frac{1}{3}x - 5$
- ✓ B. The inverse of f is $g(x) = \frac{1}{3}(x - 5)$
- C. The inverse of f is $g(x) = \frac{1}{3x+5}$
- D. f has no inverse

Find the inverse

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Let $f(x) = x^2$ with domain $\mathbb{R}$ and with codomain $[0, \infty)$.

- A. The inverse of f is $g(x) = \sqrt{x}$
- B. The inverse of f is $g(x) = \pm\sqrt{x}$
- C. The inverse of f is $g(x) = \sqrt{|x|}$
- ✓ D. f has no inverse

Find the inverse

Let $f(x) = \sqrt{x}$ with domain $[0, \infty)$ and with codomain $\mathbb{R}$.

- A. The inverse of f is $g(x) = x^2$ on the domain $\mathbb{R}$
- ✓ B. The inverse of f is $g(x) = x^2$ on the domain $[0, \infty)$
- C. The inverse of f is $g(x) = x^{-\frac{1}{2}}$ on the domain $(0, \infty)$
- D. f has no inverse

Inverse functions

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Proposition:

A function f is invertible precisely if it is one-to-one.

That is:

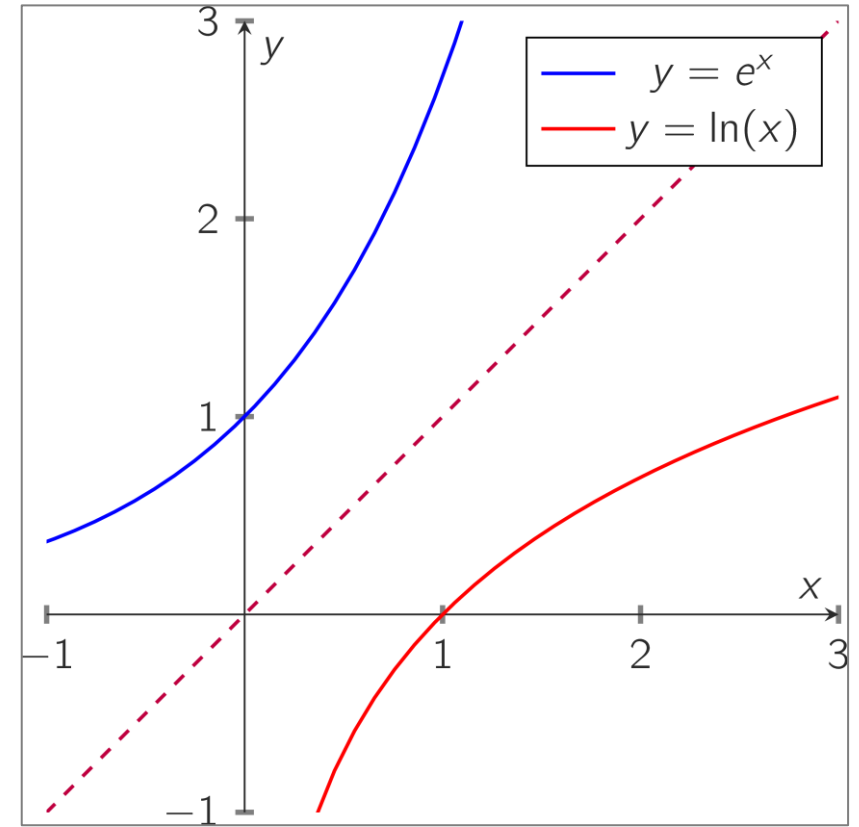
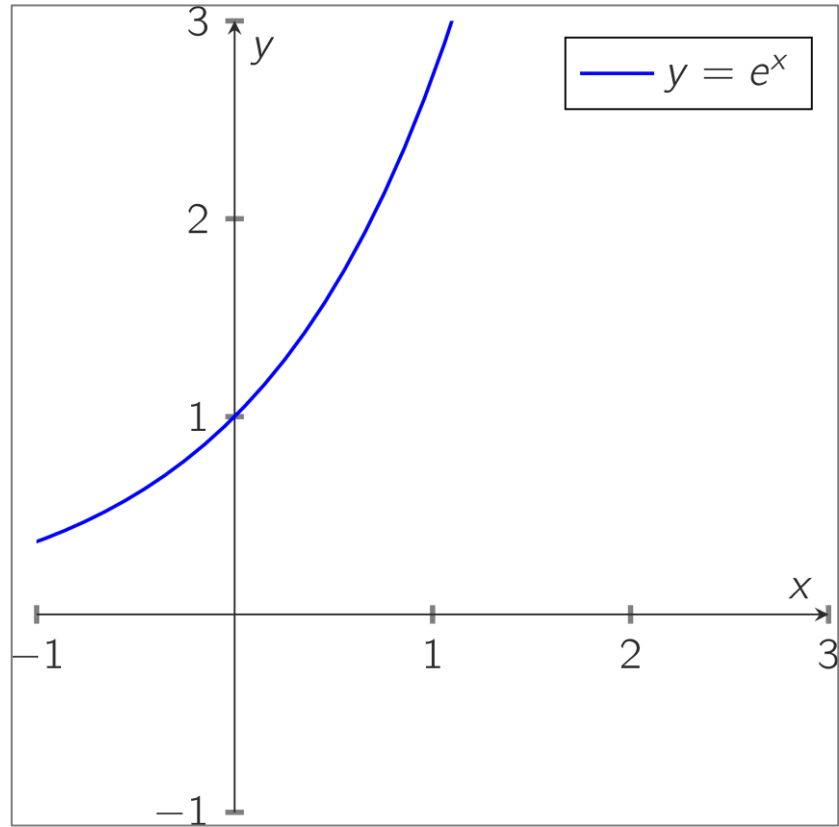
$$f(x_1) \neq f(x_2) \quad \text{for all} \quad x_1 \neq x_2$$



Inverse functions

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Suppose f is a function with inverse f^{-1} , then you get the graph of f^{-1} by reflecting the graph of f in the line $y = x$.



How to find the inverse of a function?

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The inverse of a one-to-one function can be obtained as follows:

1. start with $y = f(x)$;
2. solve this equation for x (in terms of y);
3. interchange the role of x and y , then we have: $y = f^{-1}(x)$.



Inverse trigonometric functions

Inverse trigonometric functions

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Definition:

$$\arcsin(x) = y \iff \sin(y) = x \quad \text{with} \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

Definition:

$$\arccos(x) = y \iff \cos(y) = x \quad \text{with} \quad 0 \leq y \leq \pi$$

Definition:

$$\arctan(x) = y \iff \tan(y) = x \quad \text{with} \quad -\frac{\pi}{2} < y < \frac{\pi}{2}$$



Arcsin

What is $\arcsin(-\frac{1}{2})$?

A. 0

B. π

✓ C. $-\frac{\pi}{6}$

D. $\frac{\pi}{6}$

Arctan

What is $\arctan(\tan(10))$?

- A. 10
- ✓ B. $10 - 3\pi$
- C. 0
- D. $\frac{1}{10}$

Simplifying expressions: examples

20

1. Simplify the expression $\cos(\arcsin(\frac{3}{5}))$
2. Simplify the expression $\cos(\arctan(x))$ (Stewart §1.5: example 14)



Practice

21

To be able to find the inverse of a function and simplify expression involving trigonometric and inverse trigonometric functions

Find the inverse functions for:

9th, §1.5, 25

$$f(x) = 2 + \sqrt{x+1}$$

8th, §1.5, 24

$$g(x) = \frac{e^x}{1+2e^x}$$

9th, §1.5, 10

$$h(x) = x^4 - 16$$

Prove that $\cos(\arcsin(x)) = \sqrt{1-x^2}$. 9th, §1.5, 75

Simplify the expression $\tan(\arcsin(x))$. 9th, §1.5, 76



The limit of a function

The limit of a function: intuition

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$\lim_{x \rightarrow a} f(x) = L$ means:

“ $f(x)$ approaches L as x approaches a (but x not equal to a !).”

$\lim_{x \rightarrow a^-} f(x) = L$ means:

“ $f(x)$ approaches L as x approaches a *from the left*.”

$\lim_{x \rightarrow a^+} f(x) = L$ means:

“ $f(x)$ approaches L as x approaches a *from the right*.”

Theorem: $\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^+} f(x) = L = \lim_{x \rightarrow a^-} f(x)$



Continuity

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Definition:

A function f is continuous at $x = a$ if $\lim_{x \rightarrow a} f(x) = f(a)$

The following are continuous on their maximal domain:

- x^r
- a^x and $f(x) = \log_a(x)$ for $a > 0$ and $a \neq 1$
- $\sin(x)$, $\tan(x)$, $\cos(x)$, $\arcsin(x)$, $\arccos(x)$, $\arctan(x)$
- $|x|$



Continuity

23

Definition:

A function f is left continuous at $x = a$ if $\lim_{x \rightarrow a^-} f(x) = f(a)$

The following are continuous on their maximal domain:

- x^r
- a^x and $f(x) = \log_a(x)$ for $a > 0$ and $a \neq 1$
- $\sin(x)$, $\tan(x)$, $\cos(x)$, $\arcsin(x)$, $\arccos(x)$, $\arctan(x)$
- $|x|$



Continuity

23

Definition:

A function f is right continuous at $x = a$ if $\lim_{x \rightarrow a^+} f(x) = f(a)$

The following are continuous on their maximal domain:

- x^r
- a^x and $f(x) = \log_a(x)$ for $a > 0$ and $a \neq 1$
- $\sin(x)$, $\tan(x)$, $\cos(x)$, $\arcsin(x)$, $\arccos(x)$, $\arctan(x)$
- $|x|$



Continuity

$$\text{Let } f(x) = \begin{cases} \frac{x}{|x|} & \text{if } x \neq 0 \\ a & \text{if } x = 0 \end{cases}$$

We want f to be continuous at $x = 0$.

- A. Any value of a between -1 and 1 will do
- B. a must be 0
- C. a must be either 1 or -1
- ✓ D. No value for a will do

Continuity

$$\text{Let } f(x) = \begin{cases} x + a & \text{if } x < 1 \\ 2^x & \text{if } x \geq 1 \end{cases}$$

We want f to be continuous at $x = 1$.

- A. Any value for a will do
- B. a must be 0
- ✓ C. a must be 1
- D. No value for a will do

Intermediate values

Intermediate value theorem

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Intermediate value theorem

Suppose f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.

Practice

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To be able to determine at which points a function is discontinuous

Sketch the function f and explain why it is not continuous at $a = 1$.

$$f(x) = \begin{cases} \frac{x^2 - x}{x^2 - 1} & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases} \quad 9^{\text{th}}, \S 2.5, 22$$

Determine at which point the function g is discontinuous, at which of these points it is continuous from the left or right or neither.

$$g(x) = \begin{cases} x^2 & \text{if } x < -1 \\ x & \text{if } -1 \leq x < 1 \\ \frac{1}{x} & \text{if } x \geq 1 \end{cases} \quad 9^{\text{th}}, \S 2.5, 43$$



Wrap-up and next lecture

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Practice the topics of this lecture to be able to:

- find the inverse of a one-to-one function
- simplify expressions involving (inverses of) trigonometric functions
- Investigate whether a function is continuous at a point.

Next lecture:



Limits



Stewart §2.2–2.5

$f(x)$



See you next lecture!

